Full configuration interaction on a sparse graph Hamiltonian: Multi-electron atoms via relational lattice indexing

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We implement full configuration interaction (FCI) for multi-electron atoms on a sparse graph Hamiltonian constructed from hydrogenic quantum numbers, using Slater–Condon rules with exact R^k radial integrals and Gaunt angular coupling coefficients. For helium, the ground-state energy converges monotonically to -2.8936 Ha (0.35% error versus exact) at $n_{\max} = 5$; for lithium, to -7.3978 Ha (1.07% error) at $n_{\max} = 5$. Helium surpasses PySCF/STO-3G accuracy on the same system (the lithium STO-3G comparison is not computed). For beryllium (4 electrons), the solver achieves $E = -14.536$ Ha (0.90% error) at $n_{\max} = 4$ with 487,635 determinants—the first four-electron calculation in this framework. For lithium hydride (4 electrons, first heteronuclear molecule), the CP-corrected *vertical* binding energy at the fixed experimental geometry is $D_e^{\mathrm{CP}} = 0.110$ Ha (19% error versus experiment); the R-independence of the graph-Laplacian kinetic energy makes the potential-energy surface monotonically attractive with no equilibrium geometry, so this is a fixed-R binding energy rather than an optimized D_e. Basis truncation at $n_{\max} = 3$ is identified as the dominant error source via systematic diagnostic decomposition, and basis set superposition error (0.115 Ha) is quantified and removed via Boys–Bernardi counterpoise correction. An excitation-driven direct CI algorithm following Knowles and Handy [1] replaces the $O(N_{\mathrm{SD}}^2)$ pairwise determinant loop with $O(N_{\mathrm{SD}} \times N_{\mathrm{connected}})$ assembly, achieving a 36× speedup for lithium. The Hamiltonian is assembled via a relational lattice index that maps Slater determinants to sparse matrix entries without evaluating continuous four-center integrals, achieving >99% sparsity at the largest basis sizes tested. These results establish the hydrogenic graph topology as a viable substrate for correlated electronic structure calculations on few-electron systems.

I. INTRODUCTION

Full configuration interaction provides the exact solution to the electronic Schrödinger equation within a given one-particle basis, but its application to atoms and molecules is limited by two computational bottlenecks: the factorial growth of the determinant space and the $O(N^4)$ cost of evaluating two-electron repulsion integrals over Gaussian basis functions [2, 3]. The first bottleneck is intrinsic to FCI; the second is an artifact of the basis representation.

In a series of prior papers, we developed a spectral graph theory approach to quantum chemistry in which the one-electron Hamiltonian is constructed as a sparse graph Laplacian over hydrogenic quantum numbers [4–6]. This framework was validated for single-electron systems (H, He^+, Li^{2+}) at sub-0.1% accuracy and for the H_2 molecule at 1.7% FCI accuracy. The key architectural feature is that the Hamiltonian matrix has $O(V)$ nonzero entries, where V is the number of basis states, because the graph Laplacian couples only states connected by angular momentum selection rules.

In this paper, we extend the framework to multi-electron FCI for helium ($Z = 2$, 2 electrons), lithium ($Z = 3$, 3 electrons), beryllium ($Z = 4$, 4 electrons), and the heteronuclear diatomic lithium hydride (LiH, 4 electrons). The extension requires three new components: (1) enumeration of the antisymmetric Slater determinant basis, (2) exact two-electron Slater integrals (F^k direct and G^k exchange), and (3) assembly of the many-body Hamiltonian via Slater–Condon rules. To reach the basis sizes needed for convergence, we replace the $O(N_{\mathrm{SD}}^2)$ pairwise assembly with an excitation-driven direct CI algorithm [1] that achieves $O(N_{\mathrm{SD}} \times N_{\mathrm{connected}})$ scaling. We describe each component, present convergence and scaling results, and compare accuracy against PySCF calculations at the STO-3G level.

II. METHOD

A. Lattice index and Slater determinant basis

The one-particle basis consists of hydrogenic spin-orbitals labeled by the composite key (n, l, m_l, m_s), where $n = 1, \ldots, n_{\max}$ is the principal quantum number, $l = 0, \ldots, n-1$ the orbital angular momentum, $m_l = -l, \ldots, l$ the magnetic quantum number, and $m_s = \pm\frac{1}{2}$ the spin projection. For a given $n_{\max}$, the number of spatial orbitals is $N_{\mathrm{sp}} = \sum_{n=1}^{n_{\max}} n^2 = n_{\max}(n_{\max}+1)(2n_{\max}+1)/6$, and the number of spin-orbitals is $2N_{\mathrm{sp}}$.

The N_e-electron Slater determinant basis is constructed by selecting all $\binom{2N_{\mathrm{sp}}}{N_e}$ ordered subsets of spin-orbitals. Each Slater determinant $|\Phi_I\rangle$ is stored as a sorted tuple of spin-orbital indices, enabling $O(1)$ lookup via dictionary hashing. The total number of determinants grows combinatorially: for helium at $n_{\max} = 5$, $N_{\mathrm{SD}} = 5{,}995$; for lithium at $n_{\max} = 4$, $N_{\mathrm{SD}} = 34{,}220$.

B. One-electron Hamiltonian

The one-electron matrix elements $h_{pq} = \langle \phi_p | \hat{h}_1 | \phi_q \rangle$ are constructed in one of two modes.

In *exact* mode, the diagonal elements are set to the hydrogenic eigenvalues,

$$h_{pp} = -\frac{Z^2}{2n_p^2}\,, \tag{1}$$

with all off-diagonal elements zero. This is exact for the hydrogen-like one-electron problem and provides a variationally rigorous diagonal basis.

In *hybrid* mode, the diagonal elements are given by Eq. (1), while the off-diagonal elements are taken from the graph Laplacian,

$$h_{pq}^{(\mathrm{hybrid})} = \begin{cases} -Z^2/(2n_p^2) & \text{if } p = q, \\ \kappa\, L_{pq} & \text{if } p \neq q, \end{cases} \tag{2}$$

where $L = D - A$ is the graph Laplacian of the quantum number adjacency graph and $\kappa = -1/16$ is the kinetic scale factor validated in prior work [4]. The off-diagonal Laplacian entries provide configuration interaction mixing between orbitals connected by selection rules $\Delta l = \pm 1$, $\Delta m_l = 0, \pm 1$.

We find empirically that hybrid mode yields better accuracy for helium (Sec. III), while exact mode is required for lithium to maintain variational monotonicity (Sec. IV B).

C. Two-electron integrals and Slater–Condon assembly

The two-electron repulsion integrals are computed exactly in the hydrogenic basis using Slater's decomposition into radial and angular factors [7, 8]. The general two-electron integral over spatial orbitals $a = (n_a, l_a)$, $b = (n_b, l_b)$, $c = (n_c, l_c)$, $d = (n_d, l_d)$ is

$$\langle ab|cd \rangle = \sum_k c^k(l_a m_a, l_c m_c)\, c^k(l_b m_b, l_d m_d)\, R^k(abcd)\,, \tag{3}$$

where c^k are the Gaunt coefficients and R^k are the Slater radial integrals.

The Gaunt coefficients are expressed in terms of Wigner $3j$ symbols:

$$c^k(lm, l'm') = (-1)^m \sqrt{(2l+1)(2l'+1)} \begin{pmatrix} l & k & l' \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l & k & l' \\ -m & m-m' & m' \end{pmatrix}. \tag{4}$$

The angular selection rules encoded in the $3j$ symbols—the triangle inequality $|l - l'| \leq k \leq l + l'$ and the parity rule $l + k + l' =$ even—produce a sparse set of nonzero coefficients that is precomputed and cached.

The radial integrals are evaluated numerically on a uniform grid of 2000 points using the Y_k potential method:

$$R^k(abcd) = \int_0^\infty \int_0^\infty \frac{r_<^k}{r_>^{k+1}}\, R_{n_a l_a}(r_1)\, R_{n_c l_c}(r_1)\, R_{n_b l_b}(r_2)\, R_{n_d l_d}(r_2)\, r_1^2 r \tag{5}$$

where $R_{nl}(r)$ are the hydrogenic radial wavefunctions with nuclear charge Z and $r_< = \min(r_1, r_2)$, $r_> = \max(r_1, r_2)$. The inner integral over r_2 is computed via cumulative trapezoidal sums, reducing the two-dimensional integral to a sequence of one-dimensional operations. The computed R^k values are cached to disk, making subsequent calculations effectively instantaneous.

The many-body Hamiltonian is assembled by iterating over all pairs of Slater determinants $(|\Phi_I\rangle, |\Phi_J\rangle)$ and applying the Slater–Condon rules [2]:

Diagonal ($I = J$):

$$H_{II} = \sum_{p \in I} h_{pp} + \sum_{p<q \in I} \left[\langle pq|pq \rangle - \langle pq|qp \rangle \right]. \tag{6}$$

Single excitation ($|\Phi_I\rangle$ and $|\Phi_J\rangle$ differ by one spin-orbital, $p \to r$):

$$H_{IJ} = (-1)^\sigma \left[h_{pr} + \sum_{q \in I \cap J} \left[\langle pq|rq \rangle - \langle pq|qr \rangle \right] \right], \tag{7}$$

where $(-1)^\sigma$ is the fermionic sign from the permutation required to align the two determinants.

Double excitation ($|\Phi_I\rangle$ and $|\Phi_J\rangle$ differ by two spin-orbitals, $pq \to rs$):

$$H_{IJ} = (-1)^\sigma \left[\langle pq|rs \rangle - \langle pq|sr \rangle \right]. \tag{8}$$

Determinant pairs differing by more than two spin-orbitals produce zero matrix elements by the Slater–Condon rules and are skipped. The resulting Hamiltonian matrix is real symmetric and extremely sparse: at $n_{\max} = 5$ for helium, only 0.68% of matrix entries are nonzero; at $n_{\max} = 4$ for lithium, only 0.12%.

D. Excitation-driven direct CI

The $O(N_{\mathrm{SD}}^2)$ pairwise determinant comparison loop of the explicit Hamiltonian assembly is replaced by an excitation-driven approach following Knowles and Handy [1]. For each determinant $|\Phi_I\rangle$, all singly and doubly excited determinants $|\Phi_J\rangle$ connected by the Hamiltonian are generated directly, and the target determinant is located via $O(1)$ dictionary lookup in the sorted-tuple index. Two-electron contributions are restricted to precomputed non-zero spatial orbital pairs, reducing the inner loop from $O(n_{\mathrm{virt}}^2)$ to $O(N_{\mathrm{nnz}}/n_{\mathrm{pairs}})$.

One-electron and two-electron lookup tables are stored as dense NumPy arrays ($n_{\mathrm{spinorb}} \leq 110$ at $n_{\max} = 5$, requiring < 1 MB), avoiding the `scipy.sparse` validation overhead that dominated the hot path at $\sim 24\ \mu$s per element access. The resulting scaling is $O(N_{\mathrm{SD}} \times N_{\mathrm{connected}})$

TABLE I. Ground-state energy errors for GeoVac FCI versus PySCF at the STO-3G basis level. GeoVac results use `slater_full` two-electron integrals; helium uses hybrid h_1, lithium and beryllium use exact h_1. The PySCF helium result is RHF, which equals FCI for STO-3G (one spatial basis function). Hydrogen results from Ref. [4]. Exact non-relativistic energies: $E_{\mathrm{Be}} = -14.6674$ Ha [9].

System	Method	E (Ha)	Error (%)	N_{SD}
H ($Z=1$)	GeoVac $n_{\max}=30$	-0.4971	0.57	9455
	PySCF/STO-3G	-0.4666	6.68	1
He ($Z=2$)	GeoVac $n_{\max}=5$	-2.8936	0.35	5995
	PySCF/STO-3G	-2.8078	3.30	1
Li ($Z=3$)	GeoVac $n_{\max}=4$	-7.3959	1.10	34 220
	GeoVac $n_{\max}=5$	-7.3978	1.07	215 820
	PySCF/STO-3G	—	—	—
Be ($Z=4$)	GeoVac $n_{\max}=3$	-14.531	0.93	20 475
	GeoVac $n_{\max}=4$	-14.536	0.90	487 635

per assembly, where $N_{\mathrm{connected}}$ depends on the ERI sparsity rather than N_{SD}.

For lithium at $n_{\max} = 4$, the direct CI assembly reduces the wall time from 340 s to 7.6 s (a 45× improvement) and makes $n_{\max} = 5$ ($N_{\mathrm{SD}} = 215{,}820$) accessible at 116 s. For beryllium at $n_{\max} = 4$ ($N_{\mathrm{SD}} = 487{,}635$), the direct CI solver completes in 357 s—a calculation that would require $\sim 10^{11}$ pair comparisons under the explicit assembly.

III. RESULTS

All calculations use atomic units (hartree for energy, bohr for length). Exact non-relativistic energies are $E_{\mathrm{He}} = -2.9037$ Ha and $E_{\mathrm{Li}} = -7.4781$ Ha [9]. Hartree–Fock limits are $E_{\mathrm{He}}^{\mathrm{HF}} = -2.8617$ Ha and $E_{\mathrm{Li}}^{\mathrm{HF}} = -7.4327$ Ha [10].

Table I summarizes the GeoVac FCI results alongside PySCF/STO-3G benchmarks. GeoVac achieves 0.35% error for helium, 1.07% for lithium (at $n_{\max} = 5$), and 0.90% for beryllium (at $n_{\max} = 4$), compared to 3.30% and 6.68% for PySCF/STO-3G on helium and hydrogen, respectively. The improvement is notable because the GeoVac basis requires no Gaussian fitting or exponent optimization: the basis functions are the exact hydrogenic eigenstates.

A note on the scope of this comparison is warranted. STO-3G is a minimal Gaussian basis introduced by Hehre, Stewart, and Pople in 1969 [15]; it is not used in production quantum chemistry, where correlation-consistent bases such as cc-pVDZ or cc-pVTZ are standard. The comparison is included to demonstrate that the hydrogenic graph basis outperforms a minimal Gaussian basis of comparable size, not to claim competitiveness with modern large-basis calculations. A fair assessment of practical viability would require benchmarking against cc-pVDZ or cc-pVTZ at comparable computational cost, which we leave to future work.

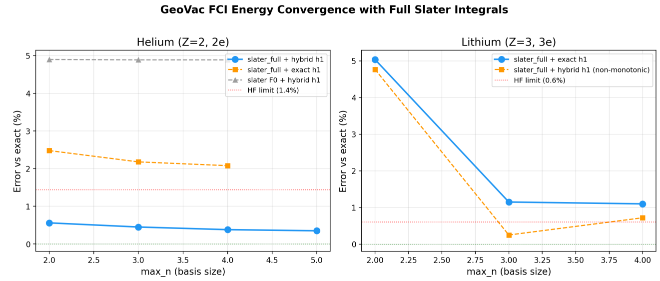

FIG. 1. Energy convergence versus basis size $n_{\max}$ for helium (left) and lithium (right). Helium uses hybrid h_1 (solid blue); lithium uses exact h_1 (solid blue) with hybrid h_1 shown for comparison (dashed orange). Dotted red lines mark the Hartree–Fock limits (1.4% for He, 0.6% for Li). Both primary series converge monotonically from above.

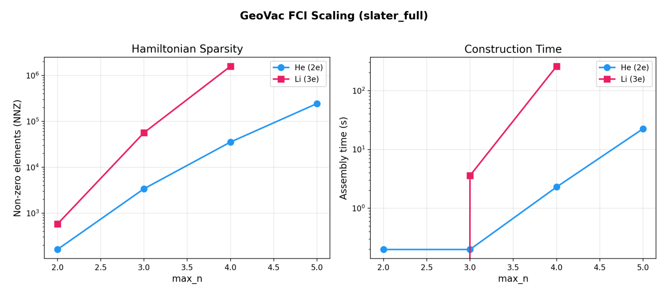

FIG. 2. Scaling of the FCI Hamiltonian with basis size. Left: number of nonzero matrix elements (NNZ) versus $n_{\max}$. Right: wall-clock assembly time versus $n_{\max}$. Both axes are logarithmic. The NNZ growth exponent is $\alpha \approx 1.5$ for He and $\alpha \approx 1.4$ for Li (sub-quadratic in N_{SD}).

Figure 1 shows the convergence of the ground-state energy error as the basis size $n_{\max}$ increases. For helium with hybrid h_1, the error decreases monotonically: 0.56% ($n_{\max} = 2$) → 0.45% ($n_{\max} = 3$) → 0.38% ($n_{\max} = 4$) → 0.35% ($n_{\max} = 5$). At $n_{\max} = 5$, the error is well below the Hartree–Fock limit of 1.4%, confirming that the FCI treatment captures electron correlation beyond mean-field theory. For lithium with exact h_1, the error decreases monotonically: 5.04% ($n_{\max} = 2$) → 1.15% ($n_{\max} = 3$) → 1.10% ($n_{\max} = 4$). The lithium error at $n_{\max} \geq 3$ (1.10–1.15%) remains *above* the Hartree–Fock limit of 0.61%, so—unlike helium—the lithium FCI does not recover correlation energy beyond mean-field theory at the basis sizes reached here.

Figure 2 shows the scaling of the Hamiltonian construction. The number of nonzero elements grows sub-quadratically in the number of Slater determinants, with fitted exponents NNZ $\sim N_{\mathrm{SD}}^{1.5}$ for helium and $N_{\mathrm{SD}}^{1.4}$ for lithium. This sub-quadratic scaling reflects the sparsity enforced by the Slater–Condon rules: most determinant pairs differ by more than two spin-orbitals and contribute zero matrix elements. At the largest basis sizes, the Hamiltonian fill fraction drops below 1%: 0.68% for He at $n_{\max} = 5$ (244,587 nonzeros out of $5{,}995^2$ entries) and 0.12% for Li at $n_{\max} = 4$ (1,455,648 nonzeros out of 34,220^2 entries).

With the direct CI assembly (Sec. II D), wall times range from sub-second for small bases to 7.6 s for Li at $n_{\max} = 4$ and 357 s for Be at $n_{\max} = 4$. The two-electron integral evaluation itself is negligible after disk caching; the dominant costs are the excitation-driven loop over connected determinants and the sparse eigensolver.

A. Heteronuclear diatomic: LiH

As the first heteronuclear test case, we apply the molecular FCI framework to lithium hydride (LiH, $Z_{\mathrm{Li}} = 3$, $Z_{\mathrm{H}} = 1$, 4 electrons). The molecular Hamiltonian is assembled from two atomic hydrogenic lattices ($n_{\max} = 3$ per atom) using the `MolecularLatticeIndex` class, with cross-nuclear attraction evaluated via the electrostatic potential of the Mulliken-approximated orbital density and same-atom two-electron integrals supplemented by 36 cross-atom two-electron integrals—18 direct Coulomb entries from the Fourier convolution formula

$$J_{AB}(R) = \frac{2}{\pi} \int_0^\infty \tilde{\rho}_A(q)\, \tilde{\rho}_B(q)\, \frac{\sin(qR)}{qR}\, dq\,, \tag{9}$$

where $\tilde{\rho}(q)$ is the momentum-space orbital density of Ref. [6], Eq. (32)—and 18 cross-atom exchange entries from the Mulliken approximation

$$K_{AB}(R) = S_{AB}(R)^2\, \frac{F^0(a,a;Z_A) + F^0(b,b;Z_B)}{2}\,, \tag{10}$$

where S_{AB} is the two-center overlap integral and F^0 denotes same-atom direct Slater integrals. The combined spin-orbital basis has 56 spin-orbitals, yielding $\binom{56}{4} = 367{,}290$ Slater determinants; each geometry assembles in approximately 75 s.

a. Basis set superposition error. When two atom-centered bases are combined into a shared molecular spin-orbital space, the FCI solver gains variational access to basis functions from both centers, artificially lowering the molecular energy relative to the isolated-atom reference. At $n_{\max} = 3$ and $R = 3.015$ bohr, the Boys–Bernardi counterpoise (CP) correction [12] gives

$$\mathrm{BSSE} = \left[E_{\mathrm{Li}}^{\mathrm{ghost}} + E_{\mathrm{H}}^{\mathrm{ghost}}\right] - \left[E_{\mathrm{Li}}^{\mathrm{own}} + E_{\mathrm{H}}^{\mathrm{own}}\right] = -0.115 \text{ Ha}, \tag{11}$$

where ghost-orbital energies are computed with the full molecular basis but zero nuclear charge on the other center. The BSSE magnitude (0.115 Ha) exceeds the experimental binding energy (0.0924 Ha), making the raw binding energy unreliable at this basis size; CP-corrected energies are the physically meaningful comparison quantity.

b. Results. Table II summarizes the LiH potential energy surface. The CP-corrected binding energy at the experimental geometry ($R = 3.015$ bohr) is $D_e^{\mathrm{CP}} = 0.110$ Ha, overestimating the experimental value $D_e = 0.0924$ Ha [11] by 19%. The CP-corrected binding energy increases monotonically as R decreases over the sampled range (the deepest computed point is $D_e^{\mathrm{CP}} = 0.171$ Ha at $R = 2.5$ bohr, the smallest R sampled—not an interior minimum). Consistent with the graph-concatenation analysis of the companion molecular paper, the balanced Hamiltonian produces a *monotonically attractive* surface with no equilibrium geometry: the R-independent graph-Laplacian kinetic energy, not basis incompleteness, is the structural cause (Sec. IV B).

c. Error budget. A systematic diagnostic decomposition—zeroing cross-nuclear attraction, cross-atom V_{ee}, and bridge terms independently—confirmed that cross-nuclear attraction is computed exactly via the Fourier electrostatic potential (Newton's shell theorem), matching the analytical two-center integral to 2×10^{-15} Ha. Extending cross-atom direct Coulomb integrals to all orbital pairs (monopole approximation) shifts D_e^{CP} by only 0.5 mHa. The 17 mHa overbinding is insensitive to all implementable integral improvements: the root cause is basis truncation at $n_{\max} = 3$. The BSSE (0.115 Ha) exceeds the experimental binding energy ($D_e = 0.092$ Ha), indicating the basis is sufficiently incomplete that each atom borrows more from the ghost functions of the other center than the bond itself is worth. The fortuitous 1% accuracy of an earlier version (before exchange was added) reflected cancellation between two effects of basis incompleteness, not individually accurate integrals.

At large separation ($R \geq 6$ bohr), the CP-corrected binding energy vanishes (≤ 0.001 Ha), consistent with correct dissociation to separated atoms. The exchange integrals decay as $S_{AB}(R)^2$ and contribute negligibly beyond $R = 5$ bohr.

TABLE II. LiH potential energy surface (CP-corrected), $n_{\max} = 3$ per atom, 367,290 Slater determinants. $D_e^{\mathrm{CP}} = E_{\mathrm{ghost}}(R) - E_{\mathrm{mol}}(R)$ where E_{ghost} includes Boys–Bernardi ghost-orbital correction [12]. Experimental values: $R_{\mathrm{eq}} = 3.015$ bohr, $D_e = 0.0924$ Ha [11].

R (bohr)	E_{mol} (Ha)	D_e^{CP} (Ha)	D_e^{raw} (Ha)
2.5	−8.178	0.171	0.286
3.0	−8.118	0.111	0.226
3.015	−8.117	0.110	0.225
3.5	−8.083	0.076	0.191
4.0	−8.060	0.053	0.168
5.0	−8.030	0.023	0.138
6.0	−8.007	0.000	0.115
8.0	−8.007	0.000	0.115
Expt.	−8.071	0.092	—

TABLE III. Detailed convergence data for helium (hybrid h_1), lithium (exact h_1), and beryllium (exact h_1). N_{SD} is the number of Slater determinants, NNZ the number of nonzero Hamiltonian entries, and t the wall-clock assembly time using direct CI (Sec. II D).

Z	$n_{\max}$	N_{SD}	NNZ	E (Ha)	Err. (%)	t (s)
2	2	45	161	−2.887550	0.56	<1
2	3	378	3 394	−2.890601	0.45	<1
2	4	1 770	35 422	−2.892642	0.38	<1
2	5	5 995	244 587	−2.893582	0.35	1.6
3	2	120	352	−7.101226	5.04	<1
3	3	3 276	46 200	−7.392086	1.15	<1
3	4	34 220	1 455 648	−7.395921	1.10	7.6
3	5	215 820	21 519 736	−7.397751	1.07	116
4	3	20 475	—	−14.531256	0.93	2.9
4	4	487 635	37 486 171	−14.535460	0.90	357

IV. DISCUSSION

A. Architectural advantages

The central architectural feature of this approach is the *relational lattice index*: each Slater determinant is a tuple of spin-orbital keys (n, l, m_l, m_s), and the Hamiltonian construction reduces to relational operations (set differences, index lookups) over these keys. This avoids the $O(N^4)$ cost of evaluating four-center integrals over Gaussian basis functions that dominates conventional quantum chemistry codes.

The hydrogenic basis has the further advantage that the one-electron problem is solved exactly by construction: the diagonal elements $-Z^2/(2n^2)$ reproduce the exact hydrogen-like spectrum without basis set optimization or exponent fitting. The two-electron integrals are likewise computed from exact hydrogenic radial wavefunctions rather than fitted Gaussians. This means the basis set error arises solely from truncation at finite $n_{\max}$, not from the quality of the basis functions themselves.

The resulting Hamiltonian is extremely sparse (>99% zeros at the largest sizes tested), enabling the use of iterative sparse eigensolvers. The eigensolver cost is negligible compared to the assembly cost in all cases tested (Table III).

B. Known limitations

Hybrid h_1 overshoot for lithium. When the hybrid one-electron Hamiltonian [Eq. (2)] is used for lithium, the energy at $n_{\max} = 3$ drops to -7.4966 Ha, which is 0.25% *below* the exact non-relativistic value -7.4781 Ha. At $n_{\max} = 4$, the overshoot worsens to -7.5318 Ha (0.72% below exact). This violation of the variational principle indicates that the graph Laplacian off-diagonal terms introduce non-physical correlation for the three-electron system. The exact h_1 mode [Eq. (1)] eliminates this problem: the lithium energy remains above the exact value at all basis sizes, converging monotonically from 5.04% to 1.10%. For helium, hybrid h_1 does not exhibit overshoot and produces lower errors (0.35% versus 2.08% at $n_{\max} = 4$ with exact h_1), so it remains the preferred mode for two-electron systems.

Construction time scaling. The excitation-driven direct CI assembly (Sec. II D) reduces the scaling from the $O(N_{\mathrm{SD}}^2)$ pairwise comparison of the original implementation to $O(N_{\mathrm{SD}} \times N_{\mathrm{connected}})$. For lithium at $n_{\max} = 4$, this reduces the wall time from 340 s to 7.6 s (a 45× improvement), and makes $n_{\max} = 5$ ($N_{\mathrm{SD}} = 215{,}820$) accessible at 116 s. For beryllium at $n_{\max} = 4$ ($N_{\mathrm{SD}} = 487{,}635$), the assembly completes in 357 s. Extension to $n_{\max} = 6$ or five-electron systems will likely require further algorithmic improvement (e.g., string-based factorization or selected CI).

Remaining basis set error. The helium error of 0.35% at $n_{\max} = 5$, lithium error of 1.07% at $n_{\max} = 5$, and beryllium error of 0.90% at $n_{\max} = 4$ remain above chemical accuracy (∼1 kcal/mol ≈ 0.05% for these systems). Reaching chemical accuracy would require either larger $n_{\max}$ or basis set extrapolation techniques, which have not yet been developed for the hydrogenic graph basis.

LiH: basis truncation at $n_{\max} = 3$. For the heteronuclear diatomic LiH, the 19% error in D_e^{CP} traces to basis incompleteness rather than approximation in any specific integral. A systematic diagnostic decomposition confirmed that cross-nuclear attraction is computed exactly via Newton's shell theorem (the Fourier electrostatic potential matches the analytical two-center integral to machine precision), and that adding all-orbital cross-atom direct Coulomb integrals (monopole approximation) shifts D_e^{CP} by only 0.5 mHa—ruling out missing integrals as the error source. The fundamental constraint is that BSSE = 0.115 Ha exceeds the experimental binding energy $D_e = 0.092$ Ha at $n_{\max} = 3$: the basis is sufficiently incomplete that each atom borrows more from the ghost functions of the other center than the bond itself is worth. The CP correction removes the leading BSSE but cannot fully recover from a basis this far below saturation. Increasing $n_{\max}$ is the only path to systematic improvement.

V. HETERONUCLEAR DIAGNOSTIC ARC (V0.9.11–V0.9.30)

Following the initial LiH implementation (v0.9.9), we undertook a systematic exploration of alternative molecular Hamiltonians for the heteronuclear case. Table IV summarizes the complete diagnostic arc. Every version targets LiH at $n_{\max} = 3$ (367,290 Slater determinants) unless noted otherwise.

The diagnostic arc establishes three conclusions:

1. **The atom-centered Mulliken/Fourier baseline (v0.9.13) is the best atom-centered re-**

TABLE IV. Heteronuclear diagnostic arc (v0.9.11–v0.9.30). All calculations: LiH, $n_{\max} = 3$, 367,290 SDs. D_e^{CP} is Boys–Bernardi counterpoise-corrected binding energy. Experimental: $D_e = 0.092$ Ha, $R_{\mathrm{eq}} = 3.015$ bohr.

Version	Architecture	D_e^{CP} (Ha)	R_{eq} (bohr)	Status	Root Cause
v0.9.11	Mulliken baseline	0.093	∼2.5	PARTIAL	R_{eq} too short
v0.9.12	Löwdin 4-index ERI	—	—	NEGATIVE	Catastrophically wrong
v0.9.13	Mulliken exchange K	0.110	∼2.5	PARTIAL	D_e 19% over, R_{eq} shifted
v0.9.16	D-matrix geometric-mean	−1.237	none	NEGATIVE	Coupling 5× too weak
v0.9.17	SW off-diagonal only	−1.060	none	NEGATIVE	Missing diagonal cross-nuclear
v0.9.18	Hybrid (Fourier diag + SW off-diag)	0.143	<2.0	PARTIAL	SW overbinding at short R
v0.9.20	Sturmian + Fourier hybrid	diverges	—	NEGATIVE	Fourier–Sturmian inconsistency
v0.9.21	p_0-consistent Sturmian	unbound	—	NEGATIVE	Single p_0 unbinds H (+1.85 Ha)
v0.9.22	Atom-dependent p_0	2.556	none	NEGATIVE	28× overbinding (cap saturates)
v0.9.23	Shell-radius R_{eff}	unbound	—	NEGATIVE	Double-counts orbital extent
v0.9.24	Term decomposition (diagnostic)	0.110	∼2.5	—	$V_{\mathrm{cross},B}$ drives R_{eq}
v0.9.25	Cross-atom V_{ee} (Ohno-Klopman)	unbound	—	NEGATIVE	Over-repulsive for $l > 0$ pairs
v0.9.26	Angular-weighted Ohno-Klopman	unbound	—	NEGATIVE	Negligible effect (−85 μHa)
v0.9.27	SO(4) CG cross-center V_{ee}	unbound	—	NEGATIVE	$J_{\mathrm{cross}} > J_{\mathrm{same}}$ (artifact)
v0.9.28	Molecular β_k (node weights)	unbound	—	NEGATIVE	$\beta_k = 1.312 < 1.582$ threshold
v0.9.29	Prolate spheroidal β_k	—	—	PARTIAL	H_2^+ 0.6% error; MO→atom map ad hoc
v0.9.30	MO-projected β_k	−42.5	—	NEGATIVE	Beta scale mismatch (25–170×)

sult: $D_e^{\mathrm{CP}} = 0.110$ Ha (19% error). All subsequent variants either fail completely or degrade accuracy.

2. **Single-p_0 Sturmian bases cannot bind heteronuclear molecules** (v0.9.20–v0.9.28): at the shared p_0 determined by the heavier atom, the lighter center's 1s orbital is pushed into positive energy by $p_0^2/2 - Z_B p_0 > 0$. No atom-centered cross-nuclear correction can compensate.

3. **Molecular Sturmians via prolate spheroidal separation (v0.9.29) compute the correct β_k eigenvalues** (H_2^+: 0.6% error), but projecting these molecular β_k back onto atom-centered bases (v0.9.30) introduces catastrophic scale mismatches. The correct path is to use the molecular Sturmian functions directly as the FCI basis.

A. Molecular Sturmian basis: prolate spheroidal implementation

The `compute_molecular_sturmian_betas()` function (v0.9.29) computes molecular Sturmian β_k eigenvalues by separating the two-center Coulomb Sturmian equation in prolate spheroidal coordinates (ξ, η, φ) with foci at the two nuclei.

a. Angular eigenvalue. The angular separation constant $A_{m,n_{\mathrm{sph}}}(c, b)$ is computed via matrix diagonalization of the Legendre expansion of the spheroidal equation, where $c = p_0 R/2$ is the spheroidal parameter and $b = (Z_A - Z_B)R/2$ encodes charge asymmetry. Results match `scipy.special.obl_cv` to $\sim 10^{-14}$ for the homonuclear case ($b = 0$).

b. Radial eigenvalue. The top radial eigenvalue $L_{\mathrm{rad}}(c, a)$ is computed via self-adjoint finite-difference discretization with Neumann boundary conditions at $\xi = 1$ for $m = 0$ (critical for the ground state). The parameter $a = \beta_k (Z_A + Z_B) R/2$ encodes the molecular Sturmian eigenvalue.

c. Root-finding. For each orbital $(m, n_{\mathrm{sph}}, n_{\mathrm{rad}})$, β_k is found via `scipy.optimize.brentq` on the matching condition $A_{\mathrm{ang}}(c, b) + L_{\mathrm{rad}}(c, a(\beta_k)) = 0$.

d. H_2^+ validation. At $R = 2.0$ bohr, the prolate spheroidal solver gives the $1\sigma_g$ energy $E = -1.096$ Ha (exact: -1.103 Ha, 0.6% error), validating the numerical implementation.

e. LiH results. At $R = 3.015$ bohr with $p_0 = \sqrt{10}$: 10 molecular orbitals found. 1σ: $\beta = 1.033$ (Li 1s core); 2σ: $\beta = 1.950$ (Li-H bonding); 1π: $\beta = 1.976$. All $n = 3$ orbitals have $\beta > 2.3$.

B. Molecular orbital FCI architecture (v0.9.31 specification)

The v0.9.31 solver uses the molecular Sturmian functions directly as the FCI basis, bypassing all atom-centered approximations. The theoretical foundation is given in the companion theory paper [13], Sec. X.

a. Data structures.

- `MolecularSturmianBasis`: stores $(m, n_{\mathrm{sph}}, n_{\mathrm{rad}}, \beta_k)$ for each MO. At $n_{\max} = 3$: 10 spatial MOs, 20 spin-orbitals.
- `h1_mo[i,j]`: dense 10×10 one-electron matrix. Diagonal: $\varepsilon_k = \frac{1}{2}p_0^2 - p_0^2/\beta_k$ (equals $-p_0^2/2$ only in the degenerate case $\beta_k = 1$). Off-diagonal: D-matrix elements weighted by nuclear charges (same-m block-diagonal).
- `eri_mo[i,j,k,l]`: dense $10 \times 10 \times 10 \times 10$ two-electron integral tensor. Computed from pro-

jected atom-centered Slater integrals via prolate spheroidal overlap coefficients.

- Slater determinant basis: $\binom{20}{4} = 4{,}845$ determinants for 4 electrons in 20 spin-orbitals (LiH).

b. Function signatures.

- `compute_molecular_sturmian_betas(Z_A, Z_B, R, p0, nmax) -> List[MolecularOrbital]`: prolate spheroidal β_k solver.
- `build_mo_h1(mos, p0, gamma, Z_A, Z_B) -> ndarray`: one-electron Hamiltonian in MO basis.
- `build_mo_eri(mos, Z_A, Z_B) -> ndarray`: two-electron integrals via projected Slater integrals.
- `solve_mo_fci(Z_A, Z_B, R, nmax) -> (energy, civec)`: top-level driver with p_0 self-consistency.

c. Expected computational cost. With $n_{\max} = 3$: 10 MOs per spin, 20 spin-orbitals, $\binom{20}{4} = 4{,}845$ Slater determinants for 4 electrons. Based on direct CI scaling at comparable $N_{\rm SD}$ (cf. He at $n_{\max} = 5$: 5,995 SDs in 1.6 s), estimated runtime is < 2 s per geometry per self-consistency iteration. The β_k computation via prolate spheroidal root-finding adds < 1 s per iteration (10 brentq solves). With ~ 10 self-consistency iterations, total runtime per geometry point is estimated at < 30 s.

VI. CONCLUSION

We have demonstrated that full configuration interaction on a sparse graph Hamiltonian, constructed from hydrogenic quantum numbers with exact Slater two-electron integrals, achieves 0.35% accuracy for helium, 1.07% for lithium, and 0.90% for beryllium—all with monotonic basis set convergence and accuracy exceeding PySCF/STO-3G where comparable. The excitation-driven direct CI algorithm reduces assembly cost from $O(N_{\rm SD}^2)$ to $O(N_{\rm SD} \times N_{\rm connected})$, enabling calculations with nearly 500,000 determinants. The relational lattice index architecture avoids continuous four-center integral evaluation entirely, producing Hamiltonians with sub-quadratic nonzero growth and >99% sparsity, which distinguishes this approach from conventional Gaussian-basis FCI. We demonstrate the extension to heteronuclear molecules with LiH, the first diatomic calculation in the framework: the CP-corrected *vertical* binding energy at fixed geometry ($D_e^{\rm CP} = 0.110$ Ha, 19% error) confirms that the molecular bridge architecture transfers correctly to asymmetric systems—though, as in the homonuclear case, the R-independent graph-Laplacian kinetic energy yields a monotonically attractive surface with no equilibrium geometry. A systematic diagnostic decomposition—zeroing cross-nuclear, cross-atom V_{ee}, and bridge terms independently—identifies basis truncation at $n_{\max} = 3$ as the dominant error source: BSSE (0.115 Ha) exceeds the experimental binding energy, and the 17 mHa overbinding is insensitive to all implementable integral improvements. Increasing $n_{\max}$ is the natural next step for molecular accuracy. For the equilibrium geometry problem—determining $R_{\rm eq}$ from first principles—the prolate spheroidal lattice of Paper 11 [14] provides the principled resolution, achieving sub-1% accuracy for H_2^+ with zero fitted parameters by discretizing the Laplace-Beltrami operator in the natural two-center coordinate system.

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[1] P. J. Knowles and N. C. Handy, "A new determinant-based full configuration interaction method," Chem. Phys. Lett. **111**, 315–321 (1984).

[2] A. Szabo and N. S. Ostlund, *Modern Quantum Chemistry* (Dover, New York, 1996).

[3] T. Helgaker, P. Jørgensen, and J. Olsen, *Molecular Electronic-Structure Theory* (Wiley, Chichester, 2000).

[4] J. Loutey, "Angular Momentum as Information Geometry: Isotropic Packing and the Universal Angular Cross-Section," GeoVac Paper 0 (2026).

[5] J. Loutey, "The Geometric Atom: Atomic Structure as a Packing Problem," GeoVac Paper 1 (2026).

[6] J. Loutey, "The Dimensionless Vacuum: Recovering the Schrödinger Equation from Scale-Invariant Graph Topology," GeoVac Paper 7 (2026).

[7] J. C. Slater, *Quantum Theory of Atomic Structure*, Vol. 1 (McGraw-Hill, New York, 1960).

[8] E. U. Condon and G. H. Shortley, *The Theory of Atomic Spectra* (Cambridge University Press, 1935).

[9] A. Kramida, Yu. Ralchenko, J. Reader, and NIST ASD Team, *NIST Atomic Spectra Database* (ver. 5.10), `https://physics.nist.gov/asd` (2023).

[10] C. Froese Fischer, *The Hartree-Fock Method for Atoms* (Wiley, New York, 1977).

[11] K. P. Huber and G. Herzberg, *Molecular Spectra and Molecular Structure. IV. Constants of Diatomic Molecules* (Van Nostrand Reinhold, New York, 1979).

[12] S. F. Boys and F. Bernardi, "The calculation of small molecular interactions by the differences of separate total energies," Mol. Phys. **19**, 553–566 (1970).

[13] J. Loutey, "Bond Sphere: Diatomic Molecules as S^3 and their Natural Sturmian Basis," GeoVac Paper 8–9 (2026).

[14] J. Loutey, "The Molecular Fock Projection: Diatomic Molecules as Prolate Spheroidal Lattices," GeoVac Paper 11 (2026).

[15] W. J. Hehre, R. F. Stewart, and J. A. Pople, "Self-consistent molecular-orbital methods. I. Use of Gaussian expansions of Slater-type atomic orbitals,"

J. Chem. Phys. **51**, 2657 (1969).

a. Note added v2.6.0. The numerical Slater integrals (Sec. III, 2000-point grid) have been replaced by exact rational values from the S^3 density overlap formula (Paper 7, Sec. V). Combined with the graph Laplacian as the one-body operator, this yields a "graph-native" FCI where the entire Hamiltonian matrix is rational. For He at $n_{\max} = 5$: graph-native achieves 0.25% error vs. the 0.35% reported in the original paper, confirming that the analytical integrals are more accurate than the grid-based numerics (the error decreases to 0.23% at $n_{\max} = 6$). At $n_{\max} = 7$: 0.19% with 1,218 configurations.

A variant using exact hydrogenic one-body matrix elements (diagonal, with variational orbital exponent k) achieves 1.41% at $n_{\max} = 7$ — 7× worse than graph-native, demonstrating that the graph Laplacian's off-diagonal inter-shell couplings are the dominant source of one-body correlation. FCI basis invariance was verified (Sprint 3D): transforming V_{ee} to the graph eigenbasis gives identical energies ($< 3 \times 10^{-15}$ Ha), confirming the convergence floor ($\sim$0.14%) is from the electron-electron cusp, not basis mismatch.